Plant Stock and Dispatch Records

This document records stock and dispatch at the plants of a manufacturer of packaged consumer goods. The daily production and daily sales-order quantities of every product are real company records (day 1 is 2023-01-01). Stock, shipments and backorders are not recorded by the company: they are computed from those records by the stock and dispatch policy stated below, applied exactly.

The document is divided into chapters titled 'Plant <id>, Period <NN>'. A period is 14 consecutive days; period 01 is days 1 to 14, period 02 is days 15 to 28, and so on. Questions refer to a chapter by that title. Days are numbered from day 1 and each day's calendar date is given next to it.

Each chapter has four sections. 'Products' lists the products dispatched by the plant, with product group, sub-group, storage location, weight per unit and opening stock. 'Daily ledger' gives, for every day of the period and every product, the units produced, the units ordered that day, the units shipped, the units backordered at the end of the day and the stock at the end of the day. 'Events' lists, in time order, the days on which the dispatch capacity was reached, products ran out of stock, orders were fully backordered and backlogs were cleared. 'Order book' gives, for the 7 days after the period, the orders already received and the production already planned; these quantities are exactly what then happened.

Stock and dispatch policy. Each plant dispatches all of its products from one storage location with one truck fleet. Every period is an independent run: on its first day each product starts with its opening stock and there are no open orders. Then, every day, in this order:

1. **Production arrives.** The day's production of each product is added to its stock.

2. **Orders join the queue.** The day's sales order of each product, if it is above zero, becomes one order (product, quantity, day placed). Orders not completely shipped on earlier days stay in the queue with their remaining quantity (they are backordered).

3. **Dispatch.** The plant has a daily dispatch capacity in kilograms, shared by all of its products. Open orders are served one at a time, oldest first; among orders placed on the same day, the one with the smaller remaining quantity goes first, then the lower product code (alphabetical order). Each order ships as many whole units as both the product's stock and the capacity still unused that day allow (a unit uses its weight in capacity). Whatever it cannot ship stays open, and the next order is served. Capacity used by one product is not available to the other products that day; unused capacity is not carried over to the next day.

4. **Records.** The ledger row of a product for a day gives: produced (units that arrived), ordered (quantity of the order placed that day, 0 if none), shipped (units shipped that day, for all of its orders together), backordered (units of all its open orders at the end of the day) and stock (units on hand at the end of the day). Stock at the end of a day = stock at the end of the previous day + produced - shipped; backordered = previous backordered + ordered - shipped.

Terms used in the Events section and in questions. A product *runs out of stock* on a day when, at the end of that day, it has no stock while some of its units are backordered; the Events section lists the first day of each such run of days. The *dispatch capacity is reached* on a day when at least one order shipped fewer units than its product's stock would have allowed because the remaining capacity was too small. An order is *fully backordered* when nothing of it ships on the day it is placed. A product's *backlog is cleared* when its backordered units return to 0. An order *ships in full* on a day when all of its units ship that day.

Constants. Opening stock of a product = 3 x its mean daily order over days 1 to 14, rounded to whole units (the same in every period; listed in each chapter's Products table). Daily dispatch capacity: Plant 1912: 34500 kg per day, Plant 1914: 192950 kg per day, Plant 2103: 407700 kg per day, Plant 2114: 300000 kg per day. Each capacity is the 80th percentile of the plant's daily ordered weight over the whole record, rounded to the nearest 50 kg.

Units. Quantities are whole units (cases); weights in kilograms (kg). Recorded quantities are rounded to whole units.

Chapters in this document (mfg_dev): Plant 1912, Period 01, Plant 1914, Period 01, Plant 2103, Period 01, Plant 2114, Period 01.

Plant 1912, Period 01

Days 1 to 14 (2023-01-01 to 2023-01-14). Plant 1912 dispatches 3 products. Daily dispatch capacity: 34500 kg.

Plant 1912, Period 01: Products

product	group	sub-group	storage location	kg per unit	opening stock
ATN02K12P	A	ATN	1230	24	2950
MAR02K12P	M	MAR	1230	24	561
MASR025K	M	MASR	1230	25	527

Plant 1912, Period 01: Daily ledger

day	date	product	produced	ordered	shipped	backordered	stock
1	2023-01-01	ATN02K12P	200	1819	873	946	2277
1	2023-01-01	MAR02K12P	100	464	464	0	197
1	2023-01-01	MASR025K	0	96	96	0	431
2	2023-01-02	ATN02K12P	500	1281	1118	1109	1659
2	2023-01-02	MAR02K12P	100	310	297	13	0
2	2023-01-02	MASR025K	100	21	21	0	510
3	2023-01-03	ATN02K12P	0	736	1401	444	258
3	2023-01-03	MAR02K12P	0	63	0	76	0
3	2023-01-03	MASR025K	0	35	35	0	475
4	2023-01-04	ATN02K12P	1700	523	967	0	991
4	2023-01-04	MAR02K12P	0	65	0	141	0
4	2023-01-04	MASR025K	5	46	46	0	434
5	2023-01-05	ATN02K12P	1000	930	930	0	1061
5	2023-01-05	MAR02K12P	0	364	0	505	0
5	2023-01-05	MASR025K	0	244	244	0	190
6	2023-01-06	ATN02K12P	800	0	0	0	1861
6	2023-01-06	MAR02K12P	300	0	300	205	0
6	2023-01-06	MASR025K	100	0	0	0	290
7	2023-01-07	ATN02K12P	400	608	608	0	1653
7	2023-01-07	MAR02K12P	7	147	7	345	0
7	2023-01-07	MASR025K	45	477	335	142	0
8	2023-01-08	ATN02K12P	0	1782	1437	345	216
8	2023-01-08	MAR02K12P	0	137	0	482	0
8	2023-01-08	MASR025K	0	231	0	373	0
9	2023-01-09	ATN02K12P	1000	925	818	452	398
9	2023-01-09	MAR02K12P	500	63	500	45	0
9	2023-01-09	MASR025K	114	34	114	293	0
10	2023-01-10	ATN02K12P	1200	1791	733	1510	865
10	2023-01-10	MAR02K12P	300	532	300	277	0
10	2023-01-10	MASR025K	510	95	388	0	122
11	2023-01-11	ATN02K12P	1000	1771	1160	2121	705
11	2023-01-11	MAR02K12P	300	300	277	300	23
11	2023-01-11	MASR025K	300	80	0	80	422
12	2023-01-12	ATN02K12P	1700	1224	1054	2291	1351
12	2023-01-12	MAR02K12P	300	161	300	161	23
12	2023-01-12	MASR025K	6	50	80	50	348
13	2023-01-13	ATN02K12P	1500	0	1224	1067	1627
13	2023-01-13	MAR02K12P	300	0	161	0	162
13	2023-01-13	MASR025K	0	0	50	0	298
14	2023-01-14	ATN02K12P	1000	378	1424	21	1203
14	2023-01-14	MAR02K12P	0	13	13	0	149
14	2023-01-14	MASR025K	52	1050	0	1050	350

Plant 1912, Period 01: Events

Day 1 (2023-01-01): dispatch capacity reached (34488.00 of 34500 kg dispatched).

Day 2 (2023-01-02): dispatch capacity reached (34485.00 of 34500 kg dispatched).

Day 2 (2023-01-02): MAR02K12P runs out of stock (13 units backordered).

Day 3 (2023-01-03): dispatch capacity reached (34499.00 of 34500 kg dispatched).

Day 3 (2023-01-03): order for MAR02K12P placed on day 3 fully backordered (63 units, none shipped).

Day 4 (2023-01-04): order for MAR02K12P placed on day 4 fully backordered (65 units, none shipped).

Day 4 (2023-01-04): backlog of ATN02K12P cleared.

Day 5 (2023-01-05): order for MAR02K12P placed on day 5 fully backordered (364 units, none shipped).

Day 7 (2023-01-07): MASR025K runs out of stock (142 units backordered).

Day 7 (2023-01-07): order for MAR02K12P placed on day 7 fully backordered (147 units, none shipped).

Day 8 (2023-01-08): dispatch capacity reached (34488.00 of 34500 kg dispatched).

Day 8 (2023-01-08): order for MAR02K12P placed on day 8 fully backordered (137 units, none shipped).

Day 8 (2023-01-08): order for MASR025K placed on day 8 fully backordered (231 units, none shipped).

Day 9 (2023-01-09): dispatch capacity reached (34482.00 of 34500 kg dispatched).

Day 9 (2023-01-09): order for MASR025K placed on day 9 fully backordered (34 units, none shipped).

Day 10 (2023-01-10): dispatch capacity reached (34492.00 of 34500 kg dispatched).

Day 10 (2023-01-10): backlog of MASR025K cleared.

Day 11 (2023-01-11): dispatch capacity reached (34488.00 of 34500 kg dispatched).

Day 11 (2023-01-11): order for ATN02K12P placed on day 11 fully backordered (1771 units, none shipped).

Day 11 (2023-01-11): order for MAR02K12P placed on day 11 fully backordered (300 units, none shipped).

Day 11 (2023-01-11): order for MASR025K placed on day 11 fully backordered (80 units, none shipped).

Day 12 (2023-01-12): dispatch capacity reached (34496.00 of 34500 kg dispatched).

Day 12 (2023-01-12): order for ATN02K12P placed on day 12 fully backordered (1224 units, none shipped).

Day 12 (2023-01-12): order for MAR02K12P placed on day 12 fully backordered (161 units, none shipped).

Day 12 (2023-01-12): order for MASR025K placed on day 12 fully backordered (50 units, none shipped).

Day 13 (2023-01-13): dispatch capacity reached (34490.00 of 34500 kg dispatched).

Day 13 (2023-01-13): backlog of MAR02K12P cleared.

Day 13 (2023-01-13): backlog of MASR025K cleared.

Day 14 (2023-01-14): dispatch capacity reached (34488.00 of 34500 kg dispatched).

Day 14 (2023-01-14): order for MASR025K placed on day 14 fully backordered (1050 units, none shipped).

Plant 1912, Period 01: Order book

Order book at the end of day 14: orders received and production planned for days 15 to 21. These quantities are exactly what then happened.

Open orders at the end of day 14 (backordered):

product	day placed	units still open
ATN02K12P	14	21
MASR025K	14	1050

Orders received and production planned for days 15 to 21:

day	date	product	planned production	ordered
15	2023-01-15	ATN02K12P	500	840
15	2023-01-15	MAR02K12P	0	47
15	2023-01-15	MASR025K	145	185
16	2023-01-16	ATN02K12P	500	2034
16	2023-01-16	MAR02K12P	0	133
16	2023-01-16	MASR025K	300	315
17	2023-01-17	ATN02K12P	0	1842
17	2023-01-17	MAR02K12P	0	536
17	2023-01-17	MASR025K	500	156
18	2023-01-18	ATN02K12P	1500	417
18	2023-01-18	MAR02K12P	302	68
18	2023-01-18	MASR025K	43	125
19	2023-01-19	ATN02K12P	1500	831
19	2023-01-19	MAR02K12P	500	153
19	2023-01-19	MASR025K	0	65
20	2023-01-20	ATN02K12P	2000	0
20	2023-01-20	MAR02K12P	0	0
20	2023-01-20	MASR025K	0	0
21	2023-01-21	ATN02K12P	1000	250
21	2023-01-21	MAR02K12P	0	46
21	2023-01-21	MASR025K	0	1

Plant 1914, Period 01

Days 1 to 14 (2023-01-01 to 2023-01-14). Plant 1914 dispatches 7 products. Daily dispatch capacity: 192950 kg.

Plant 1914, Period 01: Products

product	group	sub-group	storage location	kg per unit	opening stock
AT5X5K	A	AT	1430	25	4777
ATN01K24P	A	ATN	1430	24	12647
ATWWP001K24P	A	ATWWP	1430	24	159
MAHS025K	M	MAHS	1430	25	82
MAR01K24P	M	MAR	1430	24	2714
SE200G24P	S	SE	1430	4.8	226
SE500G24P	S	SE	1430	6	2383

Plant 1914, Period 01: Daily ledger

day	date	product	produced	ordered	shipped	backordered	stock
1	2023-01-01	AT5X5K	1500	2642	2642	0	3635
1	2023-01-01	ATN01K24P	3500	5184	3190	1994	12957
1	2023-01-01	ATWWP001K24P	0	42	42	0	117
1	2023-01-01	MAHS025K	48	25	25	0	105
1	2023-01-01	MAR01K24P	100	1607	1607	0	1207
1	2023-01-01	SE200G24P	0	225	225	0	1
1	2023-01-01	SE500G24P	1000	1507	1507	0	1876
2	2023-01-02	AT5X5K	1500	1070	1070	0	4065
2	2023-01-02	ATN01K24P	4000	4345	5738	601	11219
2	2023-01-02	ATWWP001K24P	0	37	37	0	80
2	2023-01-02	MAHS025K	0	0	0	0	105
2	2023-01-02	MAR01K24P	200	999	999	0	408
2	2023-01-02	SE200G24P	0	109	1	108	0
2	2023-01-02	SE500G24P	300	602	602	0	1574
3	2023-01-03	AT5X5K	2000	2355	2355	0	3710
3	2023-01-03	ATN01K24P	0	5891	3586	2906	7633
3	2023-01-03	ATWWP001K24P	0	51	51	0	29
3	2023-01-03	MAHS025K	0	0	0	0	105
3	2023-01-03	MAR01K24P	1200	2004	1608	396	0
3	2023-01-03	SE200G24P	0	100	0	208	0
3	2023-01-03	SE500G24P	300	1363	1363	0	511
4	2023-01-04	AT5X5K	2000	910	910	0	4800
4	2023-01-04	ATN01K24P	3000	4699	5750	1855	4883
4	2023-01-04	ATWWP001K24P	0	49	29	20	0
4	2023-01-04	MAHS025K	0	2	2	0	103
4	2023-01-04	MAR01K24P	1400	761	1157	0	243
4	2023-01-04	SE200G24P	0	130	0	338	0
4	2023-01-04	SE500G24P	100	757	611	146	0
5	2023-01-05	AT5X5K	2000	3504	3504	0	3296
5	2023-01-05	ATN01K24P	5000	3552	2972	2435	6911
5	2023-01-05	ATWWP001K24P	0	18	0	38	0
5	2023-01-05	MAHS025K	0	37	37	0	66
5	2023-01-05	MAR01K24P	2500	1129	1129	0	1614
5	2023-01-05	SE200G24P	0	155	0	493	0
5	2023-01-05	SE500G24P	1000	1398	1000	544	0
6	2023-01-06	AT5X5K	1000	0	0	0	4296
6	2023-01-06	ATN01K24P	5	0	2435	0	4481
6	2023-01-06	ATWWP001K24P	1000	0	38	0	962
6	2023-01-06	MAHS025K	0	0	0	0	66

day	date	product	produced	ordered	shipped	backordered	stock
6	2023-01-06	MAR01K24P	300	0	0	0	1914
6	2023-01-06	SE200G24P	0	0	0	493	0
6	2023-01-06	SE500G24P	400	0	400	144	0
7	2023-01-07	AT5X5K	1500	10	10	0	5786
7	2023-01-07	ATN01K24P	3000	324	324	0	7157
7	2023-01-07	ATWWP001K24P	883	0	0	0	1845
7	2023-01-07	MAHS025K	56	0	0	0	122
7	2023-01-07	MAR01K24P	300	81	81	0	2133
7	2023-01-07	SE200G24P	295	0	295	198	0
7	2023-01-07	SE500G24P	400	24	168	0	232
8	2023-01-08	AT5X5K	1500	3322	3322	0	3964
8	2023-01-08	ATN01K24P	3000	9714	2047	7667	8110
8	2023-01-08	ATWWP001K24P	0	101	101	0	1744
8	2023-01-08	MAHS025K	0	240	122	118	0
8	2023-01-08	MAR01K24P	350	2100	2100	0	383
8	2023-01-08	SE200G24P	102	97	102	193	0
8	2023-01-08	SE500G24P	500	1553	732	821	0
9	2023-01-09	AT5X5K	1500	2254	0	2254	5464
9	2023-01-09	ATN01K24P	6000	6696	7667	6696	6443
9	2023-01-09	ATWWP001K24P	433	80	80	0	2097
9	2023-01-09	MAHS025K	0	0	0	118	0
9	2023-01-09	MAR01K24P	800	1099	242	857	941
9	2023-01-09	SE200G24P	65	54	65	182	0
9	2023-01-09	SE500G24P	150	1163	150	1834	0
10	2023-01-10	AT5X5K	2000	2414	2254	2414	5210
10	2023-01-10	ATN01K24P	5050	7413	4553	9556	6940
10	2023-01-10	ATWWP001K24P	0	95	0	95	2097
10	2023-01-10	MAHS025K	200	30	118	30	82
10	2023-01-10	MAR01K24P	2500	1236	857	1236	2584
10	2023-01-10	SE200G24P	0	48	0	230	0
10	2023-01-10	SE500G24P	635	1002	635	2201	0
11	2023-01-11	AT5X5K	2000	1453	2414	1453	4796
11	2023-01-11	ATN01K24P	5500	4371	3987	9940	8453
11	2023-01-11	ATWWP001K24P	0	152	95	152	2002
11	2023-01-11	MAHS025K	0	25	30	25	52
11	2023-01-11	MAR01K24P	550	749	1236	749	1898
11	2023-01-11	SE200G24P	598	105	231	104	367
11	2023-01-11	SE500G24P	518	631	518	2314	0
12	2023-01-12	AT5X5K	2000	2020	1269	2204	5527
12	2023-01-12	ATN01K24P	5500	6371	5569	10742	8384
12	2023-01-12	ATWWP001K24P	0	104	152	104	1850
12	2023-01-12	MAHS025K	68	24	25	24	95
12	2023-01-12	MAR01K24P	400	785	749	785	1549
12	2023-01-12	SE200G24P	0	30	108	26	259
12	2023-01-12	SE500G24P	800	1094	800	2608	0
13	2023-01-13	AT5X5K	2000	0	2204	0	5323
13	2023-01-13	ATN01K24P	5500	0	4524	6218	9360
13	2023-01-13	ATWWP001K24P	0	0	104	0	1746
13	2023-01-13	MAHS025K	0	0	24	0	71
13	2023-01-13	MAR01K24P	1100	0	785	0	1864
13	2023-01-13	SE200G24P	0	0	26	0	233
13	2023-01-13	SE500G24P	1200	0	1200	1408	0
14	2023-01-14	AT5X5K	1500	339	339	0	6484
14	2023-01-14	ATN01K24P	5000	457	6675	0	7685
14	2023-01-14	ATWWP001K24P	0	13	13	0	1733
14	2023-01-14	MAHS025K	0	0	0	0	71
14	2023-01-14	MAR01K24P	1000	113	113	0	2751

day	date	product	produced	ordered	shipped	backordered	stock
14	2023-01-14	SE200G24P	0	0	0	0	233
14	2023-01-14	SE500G24P	1500	27	1435	0	65

Plant 1914, Period 01: Events

Day 1 (2023-01-01): dispatch capacity reached (192933.00 of 192950 kg dispatched).

Day 2 (2023-01-02): dispatch capacity reached (192942.80 of 192950 kg dispatched).

Day 2 (2023-01-02): SE200G24P runs out of stock (108 units backordered).

Day 3 (2023-01-03): dispatch capacity reached (192933.00 of 192950 kg dispatched).

Day 3 (2023-01-03): MAR01K24P runs out of stock (396 units backordered).

Day 3 (2023-01-03): order for SE200G24P placed on day 3 fully backordered (100 units, none shipped).

Day 4 (2023-01-04): dispatch capacity reached (192930.00 of 192950 kg dispatched).

Day 4 (2023-01-04): ATWWP001K24P runs out of stock (20 units backordered).

Day 4 (2023-01-04): SE500G24P runs out of stock (146 units backordered).

Day 4 (2023-01-04): order for SE200G24P placed on day 4 fully backordered (130 units, none shipped).

Day 4 (2023-01-04): backlog of MAR01K24P cleared.

Day 5 (2023-01-05): dispatch capacity reached (192949.00 of 192950 kg dispatched).

Day 5 (2023-01-05): order for ATWWP001K24P placed on day 5 fully backordered (18 units, none shipped).

Day 5 (2023-01-05): order for SE200G24P placed on day 5 fully backordered (155 units, none shipped).

Day 6 (2023-01-06): backlog of ATN01K24P cleared.

Day 6 (2023-01-06): backlog of ATWWP001K24P cleared.

Day 7 (2023-01-07): backlog of SE500G24P cleared.

Day 8 (2023-01-08): dispatch capacity reached (192933.60 of 192950 kg dispatched).

Day 8 (2023-01-08): MAHS025K runs out of stock (118 units backordered).

Day 8 (2023-01-08): SE500G24P runs out of stock (821 units backordered).

Day 8 (2023-01-08): order for SE200G24P placed on day 8 fully backordered (97 units, none shipped).

Day 9 (2023-01-09): dispatch capacity reached (192948.00 of 192950 kg dispatched).

Day 9 (2023-01-09): order for AT5X5K placed on day 9 fully backordered (2254 units, none shipped).

Day 9 (2023-01-09): order for ATN01K24P placed on day 9 fully backordered (6696 units, none shipped).

Day 9 (2023-01-09): order for SE200G24P placed on day 9 fully backordered (54 units, none shipped).

Day 9 (2023-01-09): order for SE500G24P placed on day 9 fully backordered (1163 units, none shipped).

Day 10 (2023-01-10): dispatch capacity reached (192950.00 of 192950 kg dispatched).

Day 10 (2023-01-10): order for AT5X5K placed on day 10 fully backordered (2414 units, none shipped).

Day 10 (2023-01-10): order for ATN01K24P placed on day 10 fully backordered (7413 units, none shipped).

Day 10 (2023-01-10): order for ATWWP001K24P placed on day 10 fully backordered (95 units, none shipped).

Day 10 (2023-01-10): order for MAHS025K placed on day 10 fully backordered (30 units, none shipped).

Day 10 (2023-01-10): order for MAR01K24P placed on day 10 fully backordered (1236 units, none shipped).

Day 10 (2023-01-10): order for SE200G24P placed on day 10 fully backordered (48 units, none shipped).

Day 10 (2023-01-10): order for SE500G24P placed on day 10 fully backordered (1002 units, none shipped).

Day 11 (2023-01-11): dispatch capacity reached (192948.80 of 192950 kg dispatched).

Day 11 (2023-01-11): order for AT5X5K placed on day 11 fully backordered (1453 units, none shipped).

Day 11 (2023-01-11): order for ATN01K24P placed on day 11 fully backordered (4371 units, none shipped).

Day 11 (2023-01-11): order for ATWWP001K24P placed on day 11 fully backordered (152 units, none shipped).

Day 11 (2023-01-11): order for MAHS025K placed on day 11 fully backordered (25 units, none shipped).

Day 11 (2023-01-11): order for MAR01K24P placed on day 11 fully backordered (749 units, none shipped).

Day 11 (2023-01-11): order for SE500G24P placed on day 11 fully backordered (631 units, none shipped).

Day 12 (2023-01-12): dispatch capacity reached (192948.40 of 192950 kg dispatched).

Day 12 (2023-01-12): order for AT5X5K placed on day 12 fully backordered (2020 units, none shipped).

Day 12 (2023-01-12): order for ATN01K24P placed on day 12 fully backordered (6371 units, none shipped).

Day 12 (2023-01-12): order for ATWWP001K24P placed on day 12 fully backordered (104 units, none shipped).

Day 12 (2023-01-12): order for MAHS025K placed on day 12 fully backordered (24 units, none shipped).

Day 12 (2023-01-12): order for MAR01K24P placed on day 12 fully backordered (785 units, none shipped).

Day 12 (2023-01-12): order for SE500G24P placed on day 12 fully backordered (1094 units, none shipped).

Day 13 (2023-01-13): dispatch capacity reached (192936.80 of 192950 kg dispatched).

Day 13 (2023-01-13): backlog of AT5X5K cleared.

Day 13 (2023-01-13): backlog of ATWWP001K24P cleared.

Day 13 (2023-01-13): backlog of MAHS025K cleared.

Day 13 (2023-01-13): backlog of MAR01K24P cleared.

Day 13 (2023-01-13): backlog of SE200G24P cleared.

Day 14 (2023-01-14): backlog of ATN01K24P cleared.

Day 14 (2023-01-14): backlog of SE500G24P cleared.

Plant 1914, Period 01: Order book

Order book at the end of day 14: orders received and production planned for days 15 to 21. These quantities are exactly what then happened.

There are no open orders at the end of day 14.

Orders received and production planned for days 15 to 21:

day	date	product	planned production	ordered
15	2023-01-15	AT5X5K	1000	2632
15	2023-01-15	ATN01K24P	4000	7164
15	2023-01-15	ATWWP001K24P	0	26
15	2023-01-15	MAHS025K	0	175
15	2023-01-15	MAR01K24P	525	1585
15	2023-01-15	SE200G24P	0	78
15	2023-01-15	SE500G24P	1500	1170
16	2023-01-16	AT5X5K	700	2303
16	2023-01-16	ATN01K24P	2500	4531
16	2023-01-16	ATWWP001K24P	0	108
16	2023-01-16	MAHS025K	150	10
16	2023-01-16	MAR01K24P	120	795
16	2023-01-16	SE200G24P	0	45
16	2023-01-16	SE500G24P	1500	631
17	2023-01-17	AT5X5K	1000	1746

day	date	product	planned production	ordered
17	2023-01-17	ATN01K24P	2000	3231
17	2023-01-17	ATWWP001K24P	0	73
17	2023-01-17	MAHS025K	0	17
17	2023-01-17	MAR01K24P	850	551
17	2023-01-17	SE200G24P	0	120
17	2023-01-17	SE500G24P	1400	452
18	2023-01-18	AT5X5K	2000	1697
18	2023-01-18	ATN01K24P	4000	4901
18	2023-01-18	ATWWP001K24P	0	71
18	2023-01-18	MAHS025K	0	20
18	2023-01-18	MAR01K24P	920	631
18	2023-01-18	SE200G24P	6	40
18	2023-01-18	SE500G24P	1200	384
19	2023-01-19	AT5X5K	2500	1457
19	2023-01-19	ATN01K24P	5000	5147
19	2023-01-19	ATWWP001K24P	0	107
19	2023-01-19	MAHS025K	99	50
19	2023-01-19	MAR01K24P	800	837
19	2023-01-19	SE200G24P	0	130
19	2023-01-19	SE500G24P	1200	1218
20	2023-01-20	AT5X5K	3000	0
20	2023-01-20	ATN01K24P	5000	0
20	2023-01-20	ATWWP001K24P	0	0
20	2023-01-20	MAHS025K	0	0
20	2023-01-20	MAR01K24P	200	0
20	2023-01-20	SE200G24P	450	0
20	2023-01-20	SE500G24P	800	0
21	2023-01-21	AT5X5K	500	507
21	2023-01-21	ATN01K24P	3000	229
21	2023-01-21	ATWWP001K24P	0	0
21	2023-01-21	MAHS025K	86	0
21	2023-01-21	MAR01K24P	500	53
21	2023-01-21	SE200G24P	200	0
21	2023-01-21	SE500G24P	1000	53

Plant 2103, Period 01

Days 1 to 14 (2023-01-01 to 2023-01-14). Plant 2103 dispatches 13 products. Daily dispatch capacity: 407700 kg.

Plant 2103, Period 01: Products

product	group	sub-group	storage location	kg per unit	opening stock
POP001L12P-A	P	POP	330	10.92	2849
POP001L12P-B	P	POP	330	10.92	903
POP002L09P	P	POP	330	16.38	228
POP005L04P	P	POP	330	18.2	108
POP500M24P	P	POP	330	10.92	507
POPF01L12P	P	POPF	330	10.92	725
POV001L24P	P	POV	330	10.92	2687
POV002L09P	P	POV	330	16.38	776
POV005L04P	P	POV	330	18.2	509
POV500M24P	P	POV	330	10.92	1817
SOS001L12P	S	SOS	330	11.04	32929
SOS002L09P	S	SOS	330	16.56	19734
SOS008L02P	S	SOS	330	14.72	1135

Plant 2103, Period 01: Daily ledger

day	date	product	produced	ordered	shipped	backordered	stock
1	2023-01-01	POP001L12P-A	506	1250	1250	0	2105
1	2023-01-01	POP001L12P-B	0	703	703	0	200
1	2023-01-01	POP002L09P	0	228	228	0	0
1	2023-01-01	POP005L04P	0	83	83	0	25
1	2023-01-01	POP500M24P	0	429	429	0	78
1	2023-01-01	POPF01L12P	200	780	780	0	145
1	2023-01-01	POV001L24P	201	1829	1829	0	1059
1	2023-01-01	POV002L09P	0	573	573	0	203
1	2023-01-01	POV005L04P	0	348	348	0	161
1	2023-01-01	POV500M24P	205	1349	1349	0	673
1	2023-01-01	SOS001L12P	4016	21394	5118	16276	31827
1	2023-01-01	SOS002L09P	2034	14556	14556	0	7212
1	2023-01-01	SOS008L02P	1008	1355	1355	0	788
2	2023-01-02	POP001L12P-A	1308	1125	1125	0	2288
2	2023-01-02	POP001L12P-B	500	461	461	0	239
2	2023-01-02	POP002L09P	101	140	101	39	0
2	2023-01-02	POP005L04P	150	70	70	0	105
2	2023-01-02	POP500M24P	401	209	209	0	270
2	2023-01-02	POPF01L12P	114	213	213	0	46
2	2023-01-02	POV001L24P	804	1277	1277	0	586
2	2023-01-02	POV002L09P	305	380	380	0	128
2	2023-01-02	POV005L04P	154	160	160	0	155
2	2023-01-02	POV500M24P	603	559	559	0	717
2	2023-01-02	SOS001L12P	7000	17059	16276	17059	22551
2	2023-01-02	SOS002L09P	6000	13085	10205	2880	3007
2	2023-01-02	SOS008L02P	1014	338	338	0	1464
3	2023-01-03	POP001L12P-A	0	655	655	0	1633
3	2023-01-03	POP001L12P-B	306	340	340	0	205
3	2023-01-03	POP002L09P	0	55	0	94	0
3	2023-01-03	POP005L04P	0	25	25	0	80
3	2023-01-03	POP500M24P	300	201	201	0	369
3	2023-01-03	POPF01L12P	0	85	46	39	0

day	date	product	produced	ordered	shipped	backordered	stock
3	2023-01-03	POV001L24P	701	999	999	0	288
3	2023-01-03	POV002L09P	1	275	129	146	0
3	2023-01-03	POV005L04P	0	190	155	35	0
3	2023-01-03	POV500M24P	900	669	669	0	948
3	2023-01-03	SOS001L12P	10000	12178	21171	8066	11380
3	2023-01-03	SOS002L09P	5053	8573	8060	3393	0
3	2023-01-03	SOS008L02P	0	226	226	0	1238
4	2023-01-04	POP001L12P-A	1208	885	885	0	1956
4	2023-01-04	POP001L12P-B	800	309	309	0	696
4	2023-01-04	POP002L09P	201	149	201	42	0
4	2023-01-04	POP005L04P	0	30	30	0	50
4	2023-01-04	POP500M24P	205	95	95	0	479
4	2023-01-04	POPF01L12P	260	55	94	0	166
4	2023-01-04	POV001L24P	2000	1250	1250	0	1038
4	2023-01-04	POV002L09P	707	438	584	0	123
4	2023-01-04	POV005L04P	0	236	0	271	0
4	2023-01-04	POV500M24P	402	781	781	0	569
4	2023-01-04	SOS001L12P	10618	10920	14220	4766	7778
4	2023-01-04	SOS002L09P	15478	8487	11880	0	3598
4	2023-01-04	SOS008L02P	0	223	223	0	1015
5	2023-01-05	POP001L12P-A	1033	1631	1631	0	1358
5	2023-01-05	POP001L12P-B	508	350	350	0	854
5	2023-01-05	POP002L09P	102	123	102	63	0
5	2023-01-05	POP005L04P	100	49	49	0	101
5	2023-01-05	POP500M24P	101	210	210	0	370
5	2023-01-05	POPF01L12P	54	195	195	0	25
5	2023-01-05	POV001L24P	1300	1112	1112	0	1226
5	2023-01-05	POV002L09P	404	374	374	0	153
5	2023-01-05	POV005L04P	500	210	481	0	19
5	2023-01-05	POV500M24P	902	851	851	0	620
5	2023-01-05	SOS001L12P	15156	16931	12852	8845	10082
5	2023-01-05	SOS002L09P	16000	11923	11923	0	7675
5	2023-01-05	SOS008L02P	0	233	233	0	782
6	2023-01-06	POP001L12P-A	238	0	0	0	1596
6	2023-01-06	POP001L12P-B	1	0	0	0	855
6	2023-01-06	POP002L09P	0	0	0	63	0
6	2023-01-06	POP005L04P	0	0	0	0	101
6	2023-01-06	POP500M24P	0	0	0	0	370
6	2023-01-06	POPF01L12P	0	0	0	0	25
6	2023-01-06	POV001L24P	1	0	0	0	1227
6	2023-01-06	POV002L09P	0	0	0	0	153
6	2023-01-06	POV005L04P	0	0	0	0	19
6	2023-01-06	POV500M24P	311	0	0	0	931
6	2023-01-06	SOS001L12P	5006	0	8845	0	6243
6	2023-01-06	SOS002L09P	5015	0	0	0	12690
6	2023-01-06	SOS008L02P	0	0	0	0	782
7	2023-01-07	POP001L12P-A	800	650	650	0	1746
7	2023-01-07	POP001L12P-B	0	25	25	0	830
7	2023-01-07	POP002L09P	151	10	73	0	78
7	2023-01-07	POP005L04P	0	0	0	0	101
7	2023-01-07	POP500M24P	306	5	5	0	671
7	2023-01-07	POPF01L12P	0	0	0	0	25
7	2023-01-07	POV001L24P	0	530	530	0	697
7	2023-01-07	POV002L09P	398	170	170	0	381
7	2023-01-07	POV005L04P	0	135	19	116	0
7	2023-01-07	POV500M24P	1007	730	730	0	1208
7	2023-01-07	SOS001L12P	9314	1419	1419	0	14138

day	date	product	produced	ordered	shipped	backordered	stock
7	2023-01-07	SOS002L09P	3033	1582	1582	0	14141
7	2023-01-07	SOS008L02P	0	15	15	0	767
8	2023-01-08	POP001L12P-A	1429	1798	1798	0	1377
8	2023-01-08	POP001L12P-B	404	536	536	0	698
8	2023-01-08	POP002L09P	0	115	78	37	0
8	2023-01-08	POP005L04P	100	35	35	0	166
8	2023-01-08	POP500M24P	0	320	320	0	351
8	2023-01-08	POPF01L12P	0	614	25	589	0
8	2023-01-08	POV001L24P	1202	1082	1082	0	817
8	2023-01-08	POV002L09P	0	390	381	9	0
8	2023-01-08	POV005L04P	401	260	376	0	25
8	2023-01-08	POV500M24P	0	547	547	0	661
8	2023-01-08	SOS001L12P	8386	17243	12499	4744	10025
8	2023-01-08	SOS002L09P	0	11858	11858	0	2283
8	2023-01-08	SOS008L02P	0	1521	767	754	0
9	2023-01-09	POP001L12P-A	544	3524	1921	1603	0
9	2023-01-09	POP001L12P-B	101	512	512	0	287
9	2023-01-09	POP002L09P	0	65	0	102	0
9	2023-01-09	POP005L04P	0	90	90	0	76
9	2023-01-09	POP500M24P	0	384	351	33	0
9	2023-01-09	POPF01L12P	0	915	0	1504	0
9	2023-01-09	POV001L24P	1403	1390	1390	0	830
9	2023-01-09	POV002L09P	0	360	0	369	0
9	2023-01-09	POV005L04P	0	210	25	185	0
9	2023-01-09	POV500M24P	0	830	661	169	0
9	2023-01-09	SOS001L12P	10004	15628	20029	343	0
9	2023-01-09	SOS002L09P	0	5834	2283	3551	0
9	2023-01-09	SOS008L02P	0	142	0	896	0
10	2023-01-10	POP001L12P-A	1500	180	1500	283	0
10	2023-01-10	POP001L12P-B	0	415	287	128	0
10	2023-01-10	POP002L09P	0	54	0	156	0
10	2023-01-10	POP005L04P	0	41	41	0	35
10	2023-01-10	POP500M24P	0	233	0	266	0
10	2023-01-10	POPF01L12P	0	46	0	1550	0
10	2023-01-10	POV001L24P	0	812	812	0	18
10	2023-01-10	POV002L09P	0	95	0	464	0
10	2023-01-10	POV005L04P	301	257	301	141	0
10	2023-01-10	POV500M24P	1500	590	759	0	741
10	2023-01-10	SOS001L12P	11726	13523	11726	2140	0
10	2023-01-10	SOS002L09P	0	3948	0	7499	0
10	2023-01-10	SOS008L02P	0	292	0	1188	0
11	2023-01-11	POP001L12P-A	832	433	716	0	116
11	2023-01-11	POP001L12P-B	609	105	233	0	376
11	2023-01-11	POP002L09P	0	55	0	211	0
11	2023-01-11	POP005L04P	100	15	15	0	120
11	2023-01-11	POP500M24P	955	68	334	0	621
11	2023-01-11	POPF01L12P	493	80	493	1137	0
11	2023-01-11	POV001L24P	1502	879	879	0	641
11	2023-01-11	POV002L09P	0	241	0	705	0
11	2023-01-11	POV005L04P	604	188	329	0	275
11	2023-01-11	POV500M24P	1505	527	527	0	1719
11	2023-01-11	SOS001L12P	15000	9037	11177	0	3823
11	2023-01-11	SOS002L09P	0	4875	0	12374	0
11	2023-01-11	SOS008L02P	2000	633	1821	0	179
12	2023-01-12	POP001L12P-A	1755	1066	1066	0	805
12	2023-01-12	POP001L12P-B	1001	437	437	0	940
12	2023-01-12	POP002L09P	0	55	0	266	0

day	date	product	produced	ordered	shipped	backordered	stock
12	2023-01-12	POP005L04P	0	67	67	0	53
12	2023-01-12	POP500M24P	0	189	189	0	432
12	2023-01-12	POPF01L12P	0	385	0	1522	0
12	2023-01-12	POV001L24P	2007	1200	1200	0	1448
12	2023-01-12	POV002L09P	0	313	0	1018	0
12	2023-01-12	POV005L04P	0	155	155	0	120
12	2023-01-12	POV500M24P	0	969	969	0	750
12	2023-01-12	SOS001L12P	15012	16827	10224	6603	8611
12	2023-01-12	SOS002L09P	14800	6839	14800	4413	0
12	2023-01-12	SOS008L02P	1514	240	240	0	1453
13	2023-01-13	POP001L12P-A	0	0	0	0	805
13	2023-01-13	POP001L12P-B	0	0	0	0	940
13	2023-01-13	POP002L09P	514	0	266	0	248
13	2023-01-13	POP005L04P	0	0	0	0	53
13	2023-01-13	POP500M24P	0	0	0	0	432
13	2023-01-13	POPF01L12P	0	0	0	1522	0
13	2023-01-13	POV001L24P	805	0	0	0	2253
13	2023-01-13	POV002L09P	1504	0	1018	0	486
13	2023-01-13	POV005L04P	0	0	0	0	120
13	2023-01-13	POV500M24P	0	0	0	0	750
13	2023-01-13	SOS001L12P	5358	0	6603	0	7366
13	2023-01-13	SOS002L09P	4760	0	4413	0	347
13	2023-01-13	SOS008L02P	0	0	0	0	1453
14	2023-01-14	POP001L12P-A	800	100	100	0	1505
14	2023-01-14	POP001L12P-B	0	20	20	0	920
14	2023-01-14	POP002L09P	0	15	15	0	233
14	2023-01-14	POP005L04P	100	0	0	0	153
14	2023-01-14	POP500M24P	104	25	25	0	511
14	2023-01-14	POPF01L12P	2	15	2	1535	0
14	2023-01-14	POV001L24P	0	179	179	0	2074
14	2023-01-14	POV002L09P	0	10	10	0	476
14	2023-01-14	POV005L04P	302	25	25	0	397
14	2023-01-14	POV500M24P	1205	79	79	0	1876
14	2023-01-14	SOS001L12P	12000	1508	1508	0	17858
14	2023-01-14	SOS002L09P	8000	533	533	0	7814
14	2023-01-14	SOS008L02P	0	80	80	0	1373

Plant 2103, Period 01: Events

Day 1 (2023-01-01): dispatch capacity reached (407693.06 of 407700 kg dispatched).

Day 2 (2023-01-02): dispatch capacity reached (407698.46 of 407700 kg dispatched).

Day 2 (2023-01-02): POP002L09P runs out of stock (39 units backordered).

Day 2 (2023-01-02): order for SOS001L12P placed on day 2 fully backordered (17059 units, none shipped).

Day 3 (2023-01-03): dispatch capacity reached (407694.38 of 407700 kg dispatched).

Day 3 (2023-01-03): POPF01L12P runs out of stock (39 units backordered).

Day 3 (2023-01-03): POV002L09P runs out of stock (146 units backordered).

Day 3 (2023-01-03): POV005L04P runs out of stock (35 units backordered).

Day 3 (2023-01-03): SOS002L09P runs out of stock (3393 units backordered).

Day 3 (2023-01-03): order for POP002L09P placed on day 3 fully backordered (55 units, none shipped).

Day 4 (2023-01-04): dispatch capacity reached (407689.34 of 407700 kg dispatched).

Day 4 (2023-01-04): order for POV005L04P placed on day 4 fully backordered (236 units, none shipped).

Day 4 (2023-01-04): backlog of POPF01L12P cleared.
Day 4 (2023-01-04): backlog of POV002L09P cleared.
Day 4 (2023-01-04): backlog of SOS002L09P cleared.
Day 5 (2023-01-05): dispatch capacity reached (407694.68 of 407700 kg dispatched).
Day 5 (2023-01-05): backlog of POV005L04P cleared.
Day 6 (2023-01-06): backlog of SOS001L12P cleared.
Day 7 (2023-01-07): POV005L04P runs out of stock (116 units backordered).
Day 7 (2023-01-07): backlog of POP002L09P cleared.
Day 8 (2023-01-08): dispatch capacity reached (407689.66 of 407700 kg dispatched).
Day 8 (2023-01-08): POP002L09P runs out of stock (37 units backordered).
Day 8 (2023-01-08): POPF01L12P runs out of stock (589 units backordered).
Day 8 (2023-01-08): POV002L09P runs out of stock (9 units backordered).
Day 8 (2023-01-08): SOS008L02P runs out of stock (754 units backordered).
Day 8 (2023-01-08): backlog of POV005L04P cleared.
Day 9 (2023-01-09): POP001L12P-A runs out of stock (1603 units backordered).
Day 9 (2023-01-09): POP500M24P runs out of stock (33 units backordered).
Day 9 (2023-01-09): POV005L04P runs out of stock (185 units backordered).
Day 9 (2023-01-09): POV500M24P runs out of stock (169 units backordered).
Day 9 (2023-01-09): SOS001L12P runs out of stock (343 units backordered).
Day 9 (2023-01-09): SOS002L09P runs out of stock (3551 units backordered).
Day 9 (2023-01-09): order for POP002L09P placed on day 9 fully backordered (65 units, none shipped).
Day 9 (2023-01-09): order for POPF01L12P placed on day 9 fully backordered (915 units, none shipped).
Day 9 (2023-01-09): order for POV002L09P placed on day 9 fully backordered (360 units, none shipped).
Day 9 (2023-01-09): order for SOS008L02P placed on day 9 fully backordered (142 units, none shipped).
Day 10 (2023-01-10): POP001L12P-B runs out of stock (128 units backordered).
Day 10 (2023-01-10): order for POP001L12P-A placed on day 10 fully backordered (180 units, none shipped).
Day 10 (2023-01-10): order for POP002L09P placed on day 10 fully backordered (54 units, none shipped).
Day 10 (2023-01-10): order for POP500M24P placed on day 10 fully backordered (233 units, none shipped).
Day 10 (2023-01-10): order for POPF01L12P placed on day 10 fully backordered (46 units, none shipped).
Day 10 (2023-01-10): order for POV002L09P placed on day 10 fully backordered (95 units, none shipped).
Day 10 (2023-01-10): order for SOS002L09P placed on day 10 fully backordered (3948 units, none shipped).
Day 10 (2023-01-10): order for SOS008L02P placed on day 10 fully backordered (292 units, none shipped).
Day 10 (2023-01-10): backlog of POV500M24P cleared.
Day 11 (2023-01-11): order for POP002L09P placed on day 11 fully backordered (55 units, none shipped).
Day 11 (2023-01-11): order for POPF01L12P placed on day 11 fully backordered (80 units, none shipped).
Day 11 (2023-01-11): order for POV002L09P placed on day 11 fully backordered (241 units, none shipped).
Day 11 (2023-01-11): order for SOS002L09P placed on day 11 fully backordered (4875 units, none shipped).
Day 11 (2023-01-11): backlog of POP001L12P-A cleared.

Day 11 (2023-01-11): backlog of POP001L12P-B cleared.

Day 11 (2023-01-11): backlog of POP500M24P cleared.

Day 11 (2023-01-11): backlog of POV005L04P cleared.

Day 11 (2023-01-11): backlog of SOS001L12P cleared.

Day 11 (2023-01-11): backlog of SOS008L02P cleared.

Day 12 (2023-01-12): dispatch capacity reached (407696.28 of 407700 kg dispatched).

Day 12 (2023-01-12): order for POP002L09P placed on day 12 fully backordered (55 units, none shipped).

Day 12 (2023-01-12): order for POPF01L12P placed on day 12 fully backordered (385 units, none shipped).

Day 12 (2023-01-12): order for POV002L09P placed on day 12 fully backordered (313 units, none shipped).

Day 13 (2023-01-13): backlog of POP002L09P cleared.

Day 13 (2023-01-13): backlog of POV002L09P cleared.

Day 13 (2023-01-13): backlog of SOS001L12P cleared.

Day 13 (2023-01-13): backlog of SOS002L09P cleared.

Day 14 (2023-01-14): order for POPF01L12P placed on day 14 fully backordered (15 units, none shipped).

Plant 2103, Period 01: Order book

Order book at the end of day 14: orders received and production planned for days 15 to 21. These quantities are exactly what then happened.

Open orders at the end of day 14 (backordered):

product	day placed	units still open
POPF01L12P	8	94
POPF01L12P	9	915
POPF01L12P	10	46
POPF01L12P	11	80
POPF01L12P	12	385
POPF01L12P	14	15

Orders received and production planned for days 15 to 21:

day	date	product	planned production	ordered
15	2023-01-15	POP001L12P-A	976	2126
15	2023-01-15	POP001L12P-B	0	807
15	2023-01-15	POP002L09P	0	231
15	2023-01-15	POP005L04P	0	81
15	2023-01-15	POP500M24P	0	288
15	2023-01-15	POPF01L12P	500	579
15	2023-01-15	POV001L24P	0	2175
15	2023-01-15	POV002L09P	0	581
15	2023-01-15	POV005L04P	0	301
15	2023-01-15	POV500M24P	0	1558
15	2023-01-15	SOS001L12P	11897	16144
15	2023-01-15	SOS002L09P	12000	11980
15	2023-01-15	SOS008L02P	1304	1180
16	2023-01-16	POP001L12P-A	1400	455
16	2023-01-16	POP001L12P-B	101	438
16	2023-01-16	POP002L09P	0	149
16	2023-01-16	POP005L04P	0	26
16	2023-01-16	POP500M24P	100	280
16	2023-01-16	POPF01L12P	413	245

day	date	product	planned production	ordered
16	2023-01-16	POV001L24P	653	790
16	2023-01-16	POV002L09P	212	200
16	2023-01-16	POV005L04P	0	125
16	2023-01-16	POV500M24P	306	503
16	2023-01-16	SOS001L12P	11500	7141
16	2023-01-16	SOS002L09P	7010	7774
16	2023-01-16	SOS008L02P	0	244
17	2023-01-17	POP001L12P-A	1010	101
17	2023-01-17	POP001L12P-B	1000	547
17	2023-01-17	POP002L09P	304	135
17	2023-01-17	POP005L04P	100	115
17	2023-01-17	POP500M24P	301	295
17	2023-01-17	POPF01L12P	377	50
17	2023-01-17	POV001L24P	3001	533
17	2023-01-17	POV002L09P	1009	230
17	2023-01-17	POV005L04P	303	250
17	2023-01-17	POV500M24P	1508	300
17	2023-01-17	SOS001L12P	8000	7806
17	2023-01-17	SOS002L09P	6956	6453
17	2023-01-17	SOS008L02P	0	1514
18	2023-01-18	POP001L12P-A	721	1405
18	2023-01-18	POP001L12P-B	1000	449
18	2023-01-18	POP002L09P	202	54
18	2023-01-18	POP005L04P	150	76
18	2023-01-18	POP500M24P	0	135
18	2023-01-18	POPF01L12P	670	385
18	2023-01-18	POV001L24P	1501	1212
18	2023-01-18	POV002L09P	512	185
18	2023-01-18	POV005L04P	401	300
18	2023-01-18	POV500M24P	0	670
18	2023-01-18	SOS001L12P	10000	7143
18	2023-01-18	SOS002L09P	8166	3725
18	2023-01-18	SOS008L02P	0	264
19	2023-01-19	POP001L12P-A	884	640
19	2023-01-19	POP001L12P-B	0	342
19	2023-01-19	POP002L09P	0	153
19	2023-01-19	POP005L04P	0	54
19	2023-01-19	POP500M24P	0	228
19	2023-01-19	POPF01L12P	280	315
19	2023-01-19	POV001L24P	0	1879
19	2023-01-19	POV002L09P	0	524
19	2023-01-19	POV005L04P	0	206
19	2023-01-19	POV500M24P	3	1078
19	2023-01-19	SOS001L12P	13002	15254
19	2023-01-19	SOS002L09P	8877	9836
19	2023-01-19	SOS008L02P	0	836
20	2023-01-20	POP001L12P-A	0	0
20	2023-01-20	POP001L12P-B	300	0
20	2023-01-20	POP002L09P	0	0
20	2023-01-20	POP005L04P	100	0
20	2023-01-20	POP500M24P	714	0
20	2023-01-20	POPF01L12P	360	0
20	2023-01-20	POV001L24P	802	0
20	2023-01-20	POV002L09P	0	0
20	2023-01-20	POV005L04P	350	0
20	2023-01-20	POV500M24P	1203	0
20	2023-01-20	SOS001L12P	3611	0

day	date	product	planned production	ordered
20	2023-01-20	SOS002L09P	3012	0
20	2023-01-20	SOS008L02P	3	0
21	2023-01-21	POP001L12P-A	0	25
21	2023-01-21	POP001L12P-B	416	48
21	2023-01-21	POP002L09P	0	0
21	2023-01-21	POP005L04P	0	0
21	2023-01-21	POP500M24P	202	0
21	2023-01-21	POPF01L12P	419	0
21	2023-01-21	POV001L24P	1501	15
21	2023-01-21	POV002L09P	0	10
21	2023-01-21	POV005L04P	0	10
21	2023-01-21	POV500M24P	800	0
21	2023-01-21	SOS001L12P	7003	2517
21	2023-01-21	SOS002L09P	3003	1442
21	2023-01-21	SOS008L02P	2002	943

Plant 2114, Period 01

Days 1 to 14 (2023-01-01 to 2023-01-14). Plant 2114 dispatches 5 products. Daily dispatch capacity: 300000 kg.

Plant 2114, Period 01: Products

product	group	sub-group	storage location	kg per unit	opening stock
SOP001L12P	S	SOP	1430	11.04	245
SOS003L04P	S	SOS	1430	11.04	3366
SOS005L04P	S	SOS	1430	18.4	24340
SOS250M48P	S	SOS	1430	11.04	1135
SOS500M24P	S	SOS	1430	11.04	12246

Plant 2114, Period 01: Daily ledger

day	date	product	produced	ordered	shipped	backordered	stock
1	2023-01-01	SOP001L12P	0	100	100	0	145
1	2023-01-01	SOS003L04P	0	1949	1949	0	1417
1	2023-01-01	SOS005L04P	3008	12594	9262	3332	18086
1	2023-01-01	SOS250M48P	0	1159	1135	24	0
1	2023-01-01	SOS500M24P	1003	8552	8552	0	4697
2	2023-01-02	SOP001L12P	0	160	145	15	0
2	2023-01-02	SOS003L04P	0	1842	1417	425	0
2	2023-01-02	SOS005L04P	6000	11677	11042	3967	13044
2	2023-01-02	SOS250M48P	0	502	0	526	0
2	2023-01-02	SOS500M24P	3500	7207	7207	0	990
3	2023-01-03	SOP001L12P	0	278	0	293	0
3	2023-01-03	SOS003L04P	0	1135	0	1560	0
3	2023-01-03	SOS005L04P	4012	8551	12518	0	4538
3	2023-01-03	SOS250M48P	0	290	0	816	0
3	2023-01-03	SOS500M24P	8500	4234	4234	0	5256
4	2023-01-04	SOP001L12P	10	58	10	341	0
4	2023-01-04	SOS003L04P	2014	967	2014	513	0
4	2023-01-04	SOS005L04P	10497	10667	10667	0	4368
4	2023-01-04	SOS250M48P	0	485	0	1301	0
4	2023-01-04	SOS500M24P	7281	4046	4046	0	8491
5	2023-01-05	SOP001L12P	400	63	400	4	0
5	2023-01-05	SOS003L04P	2022	1783	2022	274	0
5	2023-01-05	SOS005L04P	16000	14864	11036	3828	9332
5	2023-01-05	SOS250M48P	0	555	0	1856	0
5	2023-01-05	SOS500M24P	8000	6357	6357	0	10134
6	2023-01-06	SOP001L12P	0	0	0	4	0
6	2023-01-06	SOS003L04P	0	0	0	274	0
6	2023-01-06	SOS005L04P	5623	0	3828	0	11127
6	2023-01-06	SOS250M48P	0	0	0	1856	0
6	2023-01-06	SOS500M24P	1603	0	0	0	11737
7	2023-01-07	SOP001L12P	0	35	0	39	0
7	2023-01-07	SOS003L04P	0	305	0	579	0
7	2023-01-07	SOS005L04P	8001	3216	3216	0	15912
7	2023-01-07	SOS250M48P	0	114	0	1970	0
7	2023-01-07	SOS500M24P	2376	796	796	0	13317
8	2023-01-08	SOP001L12P	100	72	100	11	0
8	2023-01-08	SOS003L04P	2005	1927	2005	501	0
8	2023-01-08	SOS005L04P	10000	13298	9631	3667	16281
8	2023-01-08	SOS250M48P	1818	495	1818	647	0
8	2023-01-08	SOS500M24P	0	7199	7199	0	6118
9	2023-01-09	SOP001L12P	0	85	0	96	0

day	date	product	produced	ordered	shipped	backordered	stock
9	2023-01-09	SOS003L04P	2014	1902	2014	389	0
9	2023-01-09	SOS005L04P	6017	10587	10651	3603	11647
9	2023-01-09	SOS250M48P	2974	643	1290	0	1684
9	2023-01-09	SOS500M24P	0	6310	6118	192	0
10	2023-01-10	SOP001L12P	202	152	202	46	0
10	2023-01-10	SOS003L04P	1818	1241	1630	0	188
10	2023-01-10	SOS005L04P	5057	7137	10740	0	5964
10	2023-01-10	SOS250M48P	0	457	457	0	1227
10	2023-01-10	SOS500M24P	3552	3580	3552	220	0
11	2023-01-11	SOP001L12P	100	70	100	16	0
11	2023-01-11	SOS003L04P	2556	979	979	0	1765
11	2023-01-11	SOS005L04P	10502	7377	7377	0	9089
11	2023-01-11	SOS250M48P	0	199	199	0	1028
11	2023-01-11	SOS500M24P	4001	3487	3707	0	294
12	2023-01-12	SOP001L12P	0	64	0	80	0
12	2023-01-12	SOS003L04P	0	1567	1567	0	198
12	2023-01-12	SOS005L04P	12351	12339	12339	0	9101
12	2023-01-12	SOS250M48P	0	344	344	0	684
12	2023-01-12	SOS500M24P	8500	4635	4635	0	4159
13	2023-01-13	SOP001L12P	0	0	0	80	0
13	2023-01-13	SOS003L04P	0	0	0	0	198
13	2023-01-13	SOS005L04P	7022	0	0	0	16123
13	2023-01-13	SOS250M48P	0	0	0	0	684
13	2023-01-13	SOS500M24P	3683	0	0	0	7842
14	2023-01-14	SOP001L12P	0	4	0	84	0
14	2023-01-14	SOS003L04P	0	111	111	0	87
14	2023-01-14	SOS005L04P	6500	1281	1281	0	21342
14	2023-01-14	SOS250M48P	0	52	52	0	632
14	2023-01-14	SOS500M24P	2927	746	746	0	10023

Plant 2114, Period 01: Events

Day 1 (2023-01-01): dispatch capacity reached (299986.24 of 300000 kg dispatched).

Day 1 (2023-01-01): SOS250M48P runs out of stock (24 units backordered).

Day 2 (2023-01-02): dispatch capacity reached (299982.56 of 300000 kg dispatched).

Day 2 (2023-01-02): SOP001L12P runs out of stock (15 units backordered).

Day 2 (2023-01-02): SOS003L04P runs out of stock (425 units backordered).

Day 2 (2023-01-02): order for SOS250M48P placed on day 2 fully backordered (502 units, none shipped).

Day 3 (2023-01-03): order for SOP001L12P placed on day 3 fully backordered (278 units, none shipped).

Day 3 (2023-01-03): order for SOS003L04P placed on day 3 fully backordered (1135 units, none shipped).

Day 3 (2023-01-03): order for SOS250M48P placed on day 3 fully backordered (290 units, none shipped).

Day 3 (2023-01-03): backlog of SOS005L04P cleared.

Day 4 (2023-01-04): order for SOP001L12P placed on day 4 fully backordered (58 units, none shipped).

Day 4 (2023-01-04): order for SOS250M48P placed on day 4 fully backordered (485 units, none shipped).

Day 5 (2023-01-05): dispatch capacity reached (299982.56 of 300000 kg dispatched).

Day 5 (2023-01-05): order for SOS250M48P placed on day 5 fully backordered (555 units, none shipped).

Day 6 (2023-01-06): backlog of SOS005L04P cleared.

Day 7 (2023-01-07): order for SOP001L12P placed on day 7 fully backordered (35 units, none shipped).

Day 7 (2023-01-07): order for SOS003L04P placed on day 7 fully backordered (305 units, none shipped).

Day 7 (2023-01-07): order for SOS250M48P placed on day 7 fully backordered (114 units, none shipped).

Day 8 (2023-01-08): dispatch capacity reached (299997.28 of 300000 kg dispatched).

Day 8 (2023-01-08): order for SOS250M48P placed on day 8 fully backordered (495 units, none shipped).

Day 9 (2023-01-09): dispatch capacity reached (299997.28 of 300000 kg dispatched).

Day 9 (2023-01-09): SOS500M24P runs out of stock (192 units backordered).

Day 9 (2023-01-09): order for SOP001L12P placed on day 9 fully backordered (85 units, none shipped).

Day 9 (2023-01-09): backlog of SOS250M48P cleared.

Day 10 (2023-01-10): backlog of SOS003L04P cleared.

Day 10 (2023-01-10): backlog of SOS005L04P cleared.

Day 11 (2023-01-11): backlog of SOS500M24P cleared.

Day 12 (2023-01-12): order for SOP001L12P placed on day 12 fully backordered (64 units, none shipped).

Day 14 (2023-01-14): order for SOP001L12P placed on day 14 fully backordered (4 units, none shipped).

Plant 2114, Period 01: Order book

Order book at the end of day 14: orders received and production planned for days 15 to 21. These quantities are exactly what then happened.

Open orders at the end of day 14 (backordered):

product	day placed	units still open
SOP001L12P	11	16
SOP001L12P	12	64
SOP001L12P	14	4

Orders received and production planned for days 15 to 21:

day	date	product	planned production	ordered
15	2023-01-15	SOP001L12P	210	195
15	2023-01-15	SOS003L04P	1449	1837
15	2023-01-15	SOS005L04P	4000	12567
15	2023-01-15	SOS250M48P	2	1338
15	2023-01-15	SOS500M24P	4072	7905
16	2023-01-16	SOP001L12P	0	10
16	2023-01-16	SOS003L04P	2060	704
16	2023-01-16	SOS005L04P	6005	5958
16	2023-01-16	SOS250M48P	0	212
16	2023-01-16	SOS500M24P	5445	3143
17	2023-01-17	SOP001L12P	205	5
17	2023-01-17	SOS003L04P	2501	1112
17	2023-01-17	SOS005L04P	5343	9072
17	2023-01-17	SOS250M48P	800	267
17	2023-01-17	SOS500M24P	2502	2797
18	2023-01-18	SOP001L12P	0	430
18	2023-01-18	SOS003L04P	3046	930
18	2023-01-18	SOS005L04P	11021	5613
18	2023-01-18	SOS250M48P	3701	105
18	2023-01-18	SOS500M24P	5	3499
19	2023-01-19	SOP001L12P	202	161
19	2023-01-19	SOS003L04P	0	1265
19	2023-01-19	SOS005L04P	12913	11374
19	2023-01-19	SOS250M48P	5	527

day	date	product	planned production	ordered
19	2023-01-19	SOS500M24P	3840	6121
20	2023-01-20	SOP001L12P	0	0
20	2023-01-20	SOS003L04P	2003	0
20	2023-01-20	SOS005L04P	1957	0
20	2023-01-20	SOS250M48P	0	0
20	2023-01-20	SOS500M24P	2263	0
21	2023-01-21	SOP001L12P	200	0
21	2023-01-21	SOS003L04P	0	1217
21	2023-01-21	SOS005L04P	3002	991
21	2023-01-21	SOS250M48P	0	6
21	2023-01-21	SOS500M24P	4001	767